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Education

2011 – 2015	**Ph.D. in Statistics**
	Thammasat University, Bangkok, Thailand
2009 – 2011	**M.S. in Applied Mathematics**
	Khonkaen University, Khonkaen, Thailand
2005 – 2009	**B.Sc. in Statistics**
	Khonkaen University, Khonkaen, Thailand

Experience

May 2022 – Present	**Position: Associate Professor**
	Department of Applied Statistics, Faculty of Applied Science,
	King Mongkut's University of Technology North Bangkok
December 2017 – May 2022	**Position: Assistant Professor**
	Department of Applied Statistics, Faculty of Applied Science,
	King Mongkut's University of Technology North Bangkok
August 2015 – December 2017	**Position: Lecturer**
	Department of Applied Statistics, Faculty of Applied Science,
	King Mongkut's University of Technology North Bangkok

Research Interests

Statistical Distribution Theory, Parametric Inference, Survival Analysis and Censored Data, R Package Development, and Machine Learning

Research Group: Research Group in Statistical Learning and Inference, King Mongkut's University of Technology North Bangkok (KMUTNB)

Online Profiles & Research Identifiers

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dblp: https://dblp.org/pid/333/1474

Publications

1) Fan, Y., Phaphan, W., Jitpattanakul, A., & Mekruksavanich, S. (2026). Knowledge Distillation for Efficient Human Activity Recognition on Resource-Constrained Smartwatches. *IEEE Access*, 14, 25854–25870. https://doi.org/10.1109/ACCESS.2026.3664977
2) Phaphan, W., & Chuchuang, P. (2026). Models for Analyzing Factors and Classifying Informal Employment. *Lobachevskii Journal of Mathematics*, 46(10), 5180–5193. https://doi.org/10.1134/S1995080225611348
3) Intharasompong, S., Wongwiwat, P., & Phaphan, W. (2026). A Comparative Performance Analysis of Data Balancing Techniques and Stacked Ensemble Models for SME Credit Risk Assessment. *Lobachevskii Journal of Mathematics*, 46(10), 5143–5157. https://doi.org/10.1134/S1995080225610495
4) Phaphan, W., Khansai, N., Abdullahi, I., & Nontapa, C. (2026). Estimating Parameters for the New Mixture Nakagami and Rayleigh Distributions: Mathematical Properties and Simulations. *WSEAS Transactions on Systems*, 25, 352–369. https://doi.org/10.37394/23202.2026.25.28
5) Abdullahi, I., & Phaphan, W. (2026). A New Nakagami-Exponential Distribution: Theoretical and Statistical Properties. *WSEAS Transactions on Systems*, 25, 297–308. https://doi.org/10.37394/23202.2026.25.23
6) Budsaba, K., Abdullahi, I., & Phaphan, W. (2025). Maximum Likelihood Estimation for the Parameters of New Mixture Maxwell–Boltzmann Distribution and Its Applications. WSEAS Transactions on Mathematics, 24, 551–562. https://doi.org/10.37394/23206.2025.24.55
7) Khansai, N., Abdullahi, I., & Phaphan, W. (2025). Bayesian Analysis of the Length-Biased Nakagami Distribution under Different Loss Functions. *WSEAS Transactions on Mathematics*, 24. https://doi.org/10.37394/23206.2025.24.67
8) Khansai, N., Phaphan, W., Sirisubtawee, S., Jamboonsri, W., Sawangkaew, N., Kobmoo, N., & Koonprasert, S. (2025). Parametric Survival Functions to Measure the Effectiveness of Metarhizium neoanisopliae and Metarhizium sulphureum against the Brown Planthopper. *Lobachevskii Journal of Mathematics*, 46(4), 1597–1614. https://doi.org/10.1134/S1995080225606411
9) Mekruksavanich, S., Rojanavasu, P., Seisungsittisunti, B., & Jitpattanakul, A. (2025). Enhancing Intelligent Transportation Systems: A Deep Learning Approach for Terrain Recognition Using Vehicular Inertial Sensors. *Lobachevskii Journal of Mathematics.*
10) Budsaba, K., Tipwong, N., Manomat, S., & Phaphan, W. (2025). Total Population Forecasting in Bangkok: A Comparative Study of Time Series and Machine Learning Models. Lobachevskii Journal of Mathematics, 46(4), 1552–1566. https://doi.org/10.1134/S199508022560640X
11) Sathitvudh, S., Wongwiwat, P., & Phaphan, W. (2025). Enhancing Tourist Forecasting in Thailand's National Parks with Zero-Inflated Models. WSEAS Transactions on Environment and Development, 21, 284–292.
12) Wongwiwat, P., Chairojwattana, A., & Phaphan, W. (2025). Machine learning approaches for credit risk assessment in SMEs. *Thailand Statistician*, 23(3), 710–731.

13) Srisuradetchai, P., Somsamai, J., & Phaphan, W. (2025). Modified likelihood approach for Wald-typed interval of the shape parameter in Weibull distribution. AIMS Mathematics, 10(1), 1–20. https://doi.org/10.3934/math.2025001

14) Phaphan, W., Sathitvudh, S., Suntornsuwan, T., Budsaba, K., & Simmachan, T. (2025). Detecting automobile insurance fraud: A novel two-step strategy using effective ensemble learning techniques. Thailand Statistician, 23(1), 144–161.

15) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2025). Sensor-based hand gesture recognition using one-dimensional deep convolutional and residual bidirectional gated recurrent unit neural network. *Lobachevskii Journal of Mathematics*, 46(1), 464–480. https://doi.org/10.1134/S1995080224608166

16) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2025). Epileptic seizure detection in EEG signals via an enhanced hybrid CNN with an integrated attention mechanism. Mathematical Biosciences and Engineering, 22(1), 73–105. https://doi.org/10.3934/mbe.2025004

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18) Piladaeng, J., Promwat, D., & Phaphan W. (2024). Model Comparison for Predicting Beer Sales Volume: a Case Study of a Certain Brand of Beer. *Journal of Science Ladkrabang*, 33(1), 150-166.

19) Mekruksavanich, S., Phaphan, W., Hnoohom, N., & Jitpattanakul, A. (2024). Recognition of sports and daily activities through deep learning and convolutional block attention. *PeerJ Computer Science,* 10, e2100. https://doi.org/10.7717/peerj-cs.2100

20) Huadsri, S., & Phaphan, W. (2024). The Development of Forecasting Models for Life Insurance Data by Employing Time-series Analysis and Machine Learning Technique. *WSEAS Transactions on Mathematics*, *23*, 196-205. https://doi.org/10.37394/23206.2024.23.23

21) Srisuradetchai, P., & Phaphan, W. (2024). Using Monte-Carlo Dropout in Deep Neural Networks for Interval Forecasting of Durian Export. *WSEAS Transactions on Systems and Control*, *19*, 10-21. https://doi.org/10.37394/23203.2024.19.2

22) Srisuradetchai, P., Niyomdecha A., & Phaphan W. (2024). Wald Intervals via Profile Likelihood for the Mean of the Inverse Gaussian Distribution. *Symmetry*, *16*(1), 93. https://doi.org/10.3390/sym16010093

23) Srisuradetchai, P., & Phaphan, W. (2024). Bootstrap Intervals for the Mean of the Weighted Mixture Generalized Gamma Distribution. *Lobachevskii Journal of Mathematics*, *44*(11), 201-216. https://doi.org/10.1134/S1995080224600614

24) Phaphan, W., Sangnuch, N. & Piladaeng, J. (2023). Comparison of the Effectiveness of Regression Models for the Number of Road Accident Injuries. *Science & Technology Asia*, *28*(4), 54-66.

25) Mekruksavanich, S., Phaphan, W., Hnoohom, N., & Jitpattanakul A. (2023). Attention-Based Hybrid Deep Learning Network for Human Activity Recognition Using WiFi Channel State Information. *Applied Sciences*, *13*(15), 8884. https://doi.org/10.3390/app13158884

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27) Phaphan, W., Abdullahi, I., & Puttamat, W. (2023). Properties and Maximum Likelihood Estimation of the Novel Mixture of Fréchet Distribution. *Symmetry*, *15*(7), 1380. https://doi.org/10.3390/sym15071380

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41) Adulyasak, T., Srasri, P., Keamapiruk, U., & Phaphan, W. (2021). The impacts of forest fires in 1998-2020 on the communities and the environment in Northern Thailand by employing descriptive statistics and analysis of extreme values. *Huachiew Chalermprakiet Science and Technology Journal*, *7*(1), 31 - 42.

42) Plai-in, N., Chananet. C., & Phaphan, W. (2020). The scale predictions of multilayer PCB board by using machine learning. *Journal of Science & Technology MSU*, *39*(6), 692 - 701.

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44) Phaphan, W., & Pimpisal, A. (2020). The predictions of a daily stock price direction from the Thai news content by using natural language processing. *The Journal of Applied Science*, *19*(1), 59 - 79.

45) Pratheeparunothai B., Chananet, C., & Phaphan W. (2020). The prediction of electricity values in the process of electrolytic copper plating in PCB production through machine learning. *Huachiew Chalermprakiet Science and Technology Journal*, *6*(1), 54 - 70.

46) Phaphan, W., & Pongsart, T. (2019). Asymptotic confidence ellipses for length-biased inverse Gaussian distribution. *The Journal of King Mongkut's University of Technology North Bangkok*, *29*(2), 332 - 341.

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50) Phaphan, W. (2017). Point estimation of Birnbuam-Saunders distribution using EM-algorithm. *Thammasat International Journal of Science and Technology*, *22*(1), 109 - 115.

51) Phaphan, W., Budsaba, K., & Volodin, A. (2016). Asymptotic properties and parameter estimation based on two-sided crack distribution. *Journal of Probability and Statistical Science, 14*(2), 149 - 170.

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Conference Proceeding

1) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2026). A Comparative Analysis of Deep Learning Neural Networks for Human Activity Recognition Using Wrist-Worn Inertial Sensor. In Proceedings of 2026 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference (ECTI DAMT & NCON). IEEE.

2) Mekruksavanich, S., Liang, Q., Phaphan, W., & Jitpattanakul, A. (2026). A Comparative Analysis of Deep Learning Networks for Video-Based Human Action Recognition. In Proceedings of 2026 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference (ECTI DAMT & NCON). IEEE.

3) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2025). IMU-Based Human Fall Detection Using Deep Residual Networks for Advancing Health and Injury Prevention. In Proceedings of 2025 Research, Invention, and Innovation Congress: Innovative Electricals and Electronics (RI2C 2025). Bangkok, Thailand.

4) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2025). Skeleton-Based Physical Exercise Recognition Using Deep Residual Networks with SE Modules for Enhancing Health in Smart Cities. In Proceedings of 2025 Research, Invention, and Innovation Congress: Innovative Electricals and Electronics (RI2C 2025). Bangkok, Thailand.

5) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2025). LowResIR-Net: Lightweight residual network for ADL recognition using low-resolution infrared sensor data. *Procedia Computer Science*, 256, 1358–1366. Elsevier.

6) Mekruksavanich, S., Hnoohom, N., Phaphan, W., & Jitpattanakul, A. (2025). A comparative study of validation methods for sensor-based human activity recognition using deep learning models. In *2025 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON)* (pp. 686–691).

7) Mekruksavanich, S., Hnoohom, N., Phaphan, W., & Jitpattanakul, A. (2025). Emotion recognition using EEG signals in human brain waves and deep learning approaches. In *2025 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON)* (pp. 363–367).

8) Thepsumritporn, S., & Phaphan, W. (2025). Wald confidence interval for the shape parameter δ of three-parameter Nakagami-Rayleigh distribution. *Journal of Physics: Conference Series*, 3042(1), 012002. IOP Publishing.

9) Intharasompong, S., Jitpattanakul, A., Mekruksavanich, S., Wongwiwat, P., & Phaphan, W. (2025). Classification models and variable selection for SME credit risk assessment in balance the data set. In *2025 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON)* (pp. 493–498). IEEE.

10) Thepsumritporn, S., Jitpattanakul, A., Mekruksavanich, S., & Phaphan, W. (2025). Negative binomial regression model for predicting the number of tourists in national parks of Thailand. In *2025 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON)* (pp. 146–151). IEEE.

11) Mekruksavanich, S., Hnoohom, N., Phaphan, W., & Jitpattanakul, A. (2025). Orthopedic walker fall detection and motion classification using bidirectional gated recurrent unit neural network. In *2025 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON)* (pp. 447–452). IEEE.

12) Abdullahi, I., & Phaphan, W. (2024). Quantile regression as an alternative to ordinary least squares: A case study with life insurance data. The 7th International Conference on Applied Statistics 2024 and The 14th

National Conference on Applied Statistics and Information Technology 2024, October 24–25, Chiang Mai, Thailand, 66–78.

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14) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2024, August 8–9). Squeeze-and-excitation hybrid network for biometric identification via photoplethysmography. In 5th Research, Invention, and Innovation Congress: Innovative Electricals and Electronics (RI2C 2024) - Proceedings (pp. 72–76). Bangkok, Thailand. https://doi.org/10.1109/RI2C64012.2024.10784356

15) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2024, August 8–9). Enhancing sensor-based human activity recognition using hybrid deep learning and data augmentation. In 5th Research, Invention, and Innovation Congress: Innovative Electricals and Electronics (RI2C 2024) - Proceedings (pp. 66–71). Bangkok, Thailand. https://doi.org/10.1109/RI2C64012.2024.10784365

16) Abdullahi, I., Job, O., & Phaphan, W. (2024, August 8–9). On the properties of new mixture of gamma and Nakagami distribution. In 5th Research, Invention, and Innovation Congress: Innovative Electricals and Electronics (RI2C 2024) - Proceedings (pp. 347–354). Bangkok, Thailand. https://doi.org/10.1109/RI2C64012.2024.10784330

17) Mekruksavanich, S., Phaphan, W., Jeefoo, P., & Jitpattanakul, A. (2024, September 13–14). Enhancing chewing detection with a CNN-BiLSTM network utilizing IMU and photoplethysmography data from an earable device. In 2024 15th IEEE International Conference on Software Engineering and Service Science (ICSESS) (pp. 123–127). Changsha, China. https://doi.org/10.1109/ICSESS62520.2024.10719376

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20) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2024, July 10–12). Leveraging residual deep neural networks and multi-device sensors for heterogeneous activity recognition. In 2024 47th International Conference on Telecommunications and Signal Processing (TSP) (pp. 156-159). Prague, Czech Republic.

21) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2024, July 10–12). Deep learning networks for human knee abnormality detection based on surface EMG signals. In 2024 47th International Conference on Telecommunications and Signal Processing (TSP) (pp. 160-163). Prague, Czech Republic.

22) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2024, July 10–12). A hybrid residual CNN with channel attention mechanism for continuous user identification using wearable motion sensors. In 2024 47th

International Conference on Telecommunications and Signal Processing (TSP) (pp. 143-146). Prague, Czech Republic.

23) Mekruksavanich, S., Phaphan, W., & Jitpattanakul, A. (2024, July 10–12). Detecting face-touching gestures with smartwatches and deep learning networks. In 2024 47th International Conference on Telecommunications and Signal Processing (TSP) (pp. 249-252). Prague, Czech Republic.

24) Phaphan, W., Jitpattanakul, A., Huadsri, S., Budsaba, K., Phapan, W., & Mekruksavanich, S. (2024, March 14–16). Modeling life insurance business growth in Thailand using SARIMAX and multilayer perceptron. In 2024 16th International Conference on Computer and Automation Engineering (ICCAE) (pp. 146-151). Melbourne, Australia.

25) Mekruksavanich, S., Jantawong, P., Phaphan, W., & Jitpattanakul, A. (2024, January 31–February 3). A sensor-based deep learning approach for recognizing daily and work activities in open environments for sanitation workers. In 2024 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON) (pp. 577-581). Chiang-mai, Thailand.

26) Mekruksavanich, S., Jantawong, P., Phaphan, W., & Jitpattanakul, A. (2024, January 31–February 3). Convolutional neural network with CBAM module for fitness activity recognition using wearable IMU sensors. In 2024 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON) (pp. 567-571). Chiang-mai, Thailand.

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28) Tipwong, N., Jitpattanakul, A., Phaphan, W., Nontapa, C., & Mekruksavanich, S. (2024, January 31–February 3). Comparison model for forecasting the total population in Bangkok. In 2024 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON) (pp. 510-514). Chiang-mai, Thailand.

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30) Puttamat, W., Phaphan, W., Jitpattanakul, A., Pianpailoon, C., & Mekruksavanich, S. (2024, January 31–February 3). Comparison of the effectiveness of machine learning techniques for the number of road traffic injuries. In 2024 Joint International Conference on Digital Arts, Media and Technology with ECTI

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34) ธนกฤต นิลมัย, จีรวรรณ จันระคร, อนิรุตต์ พุฒิสันติกุล, และ วิกานดา ผาพันธ์. (2023). การพยากรณ์ปริมาณการผลิตน้ำมันภายในประเทศไทยโดยใช้การเรียนรู้ของเครื่อง. *วารสารประชุมวิชาการ: การประชุมวิชาการเสนอผลงานวิจัยระดับชาติ ครั้งที่ 11* (น. 112-125). กรุงเทพมหานคร, ประเทศไทย.

35) Phaphan, W., Ananitthi, T., Abdullahi, I., & Thepsumritporn, S. (2022). "MGGD: R Package for Mixture Generalized Gamma Distribution." In Proceedings of 2022 Research, Invention, and Innovation Congress: Innovative Electricals and Electronics (RI2C) (pp. 196-198). Bangkok, Thailand.

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Book

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